MILITARY SPECIFICATION

REFUELING UNIT, AIRCRAFT, 2000 GALLON

TYPE MC-1A

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers one type of 2000-gallon capacity aircraft refueling vehicle. The vehicle shall contain all related pumping equipment and accessories necessary for transporting and dispensing fuel to aircraft.

1.2 The tank and associated pumping and dispensing equipment are to be mounted on a military chassis.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

TT-P-485	-	Enamel, Semi-Gloss Rust-Inhibiting
TT-P-645	-	Primer, Paint, Zinc-Chromate, Alkyd Type

Military

MIL-F-00254	-	Filter/Separator, Liquid Fuel, Steel or Aluminum
MIL-N-4180	-	Nozzles, Fuel and Oil Servicing
MIL-G-5572	-	Gasoline, Aviation, Grades 80/87, 91/96, 100/130, 115/145
MIL-F-5624	-	Jet Fuel, Grades JP-4 and JP-5
MIL-N-5877	-	Nozzle, Pressure Fuel Servicing, Locking Type D-1. (As amended by WEPS Interim Amendment of 20 Sep 1965).

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MIL-H-6615 - Hose, Rubber, Gasoline with reusable couplings, Low temperature
MIL-F-8901 - Filter/Separators, Aviation and Motor Fuel, Ground and Shipboard use, Performance Requirements and Test procedures for
MIL-Q-9858 - Quality Program Requirements
MIL-C-19179 - Coupling Assembly, Quick-Disconnect, Fuel
MIL-E-22246 - Extinguisher, Fire, Dry Chemical, Portable, Cartridge Operated
MIL-H-22279 - Hose Assemblies, Rubber, Aircraft Fueling, Collapsible, By Negative Pressure
MIL-F-52308 - Filter Element, Fluid Pressure

STANDARDS

Federal

FED-STD-595 - Colors

Military

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U. S. Military Property
MIL-STD-193 - Painting Procedures, Tactical vehicles, Tracked and Wheeled
MIL-STD-33586 - Metals, Definition of Dissimilar

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NAVDOCKS Publication - P-300

(Application for copies should be addressed to the Commander, Naval Facilities Engineering Command, Memorial Drive "H" Road, Arlington, Virginia).

American Society of Mechanical Engineers, Safety Code for Industrial Power.

(Applications for copies should be addressed to the American Society of Mechanical Engineers, 485 Lexington Avenue, New York, New York).

Interstate Commerce Commission, Motor Carriers Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C.)

Underwriters' Laboratories, Inc. Publication, National Electrical Code for Hazardous Locations.

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(Applications for copies should be addressed to Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York, New York).

National Bureau of Standards, Handbook No. H-44

(Copies of Handbook No. H-44 may be obtained upon application to the Superintendent of Documents, Government Printing Office, Washington, D. C.).

3. REQUIREMENTS

3.1 General.- The refueling units covered by this specification shall be complete and ready for operation when delivered. They shall be constructed to meet all performance requirements and standards when tested as specified herein. The refueling units shall be the product of a manufacturer engaged in the fabrication and furnishing of essentially similar catalog items for commercial consumption prior to the date of invitation for bids and shall conform to the applicable requirements of the American Society of Mechanical Engineers, Safety Code for Industrial Power, and Interstate Commerce Commission.

3.1.1 Preproduction Model.- The contractor shall submit a preproduction model for examination, inspection and testing as set forth in Section 4 of this specification and such other tests deemed necessary by the cognizant inspector to determine compliance with all provisions of this specification and the contract. The preproduction model shall represent the identical quality of materials, workmanship, and design which will be used in the fabrication of the production units. The approved preproduction model shall serve as a manufacturing standard and shall be used as a guide in the inspection of production items produced under the contract. Fabrication of any production units shall not commence until the preproduction model has been approved and accepted by the Government. The approved preproduction model shall be presented for acceptance as the last unit for delivery under the contract. Examination and testing, whenever practicable, will be conducted at the contractor's plant.

3.2 Materials.- Materials used in the manufacture of the tanks and accessories shall conform to the applicable specifications specified herein. Materials covered by contractor's specifications may be used providing the specifications are equivalent to the Government's specifications and contain requirements for adequate test of material. The use of contractor's specifications shall not constitute a waiver of government inspection. When materials are used which are procured under commercially recognized non-government specifications, a certificate acceptable to the Government Inspector, showing compliance with such specifications, shall be furnished by the supplier of such material. Materials which are not covered by applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practical weight, and suitable for the purpose intended. Every effort shall be made to use only such materials as are readily available in quantity.

3.3 Protective Treatment.- The materials used in the construction of the unit shall be protected against corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification.

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3.3.1 Dissimilar Metal.- The use of dissimilar metals in intimate contact shall be held to an obsolute minimum. Dissimilar metals are defined in Drawing MS-33586.

3.4 Design and Construction.- The complete vehicle shall be a government furnished automotive chassis equipped with a 2000-gallon tank and all accessories and parts necessary to provide a complete and efficient aircraft fueling and defueling vehicle. The vehicle shall be constructed in accordance with first-class commercial practice and in accordance with applicable Interstate Commerce Commission Motor Carrier Safety Regulations. The vehicle shall be suitable for the purpose intended. The overall width of the vehicle shall not exceed 96 inches.

3.4.1 Appearance.- The assembled unit shall present a neat appearance. Sharp corners shall be avoided wherever possible. Evidence of carelessness or poor workmanship will be sufficient cause for rejection.

3.5 Chassis.- The chassis will be a 6x6, military vehicle. The designation is "Ordnance Chassis MG3A2". Unless otherwise specified in the contract the chassis will be furnished by the government.

3.5.1 Truck Muffler.- Contractor shall furnish a spark arresting muffler to replace the conventional muffler supplied with the government furnished chassis. The spark arresting muffler furnished shall have been tested in accordance with Specification MIL-A-27302 and found to meet all the requirements of said specification. The vehicle exhaust system shall be redesigned so as to locate the spark arresting muffler in the front of the vehicle underneath the right front fender with muffler discharge directed to the curb side. It shall be positioned in a manner that will permit easy accessibility for purposes of removing accumulated carbon particles. The spark arresting muffler shall not interfere with the normal operation of the wheels or other components of the vehicle chassis. Disposal of mufflers supplied with the government furnished chassis shall be as directed by the cognizant inspector.

3.5.1.1 Deep Water Fording Kit.- The government will furnish a Deep Water Fording Kit. The contractor shall modify the exhaust of the spark arresting muffler in such a way that, when required, the deep water fording kit can be installed by field personnel with simple hand tools. Provisions shall be made for stowing the deep water fording kit on the vehicle when not in use. Method of securing shall be such that the deep water fording kit will not come loose when the vehicle is used in off road service over rough terrain.

3.5.1.2 Vents.- Where vents are provided in the manufacturer's system design or where they are included in the government furnished equipment, provisions shall be made when required, for extending vent lines so that they exhaust at least 60 inches from the ground.

3.5.2 Safety Requirements.- All electrical installations and equipment which might be of an arc or spark producing nature such as electric tachometers, generators, motors, etc., installed in any compartment which contains fuel handling equipment or piping shall conform to the requirements of the National Electrical Code For Hazardous Locations, Class I, Group D, Division 2.

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3.6 Tank.-

3.6.1 Tank Body.- Tank shall be of one compartment type having a capacity of 2000 gallons at 69°F, plus at least 3 percent expansion space. The tank shall be constructed of Type 304L stainless steel. A simple segregator sump shall be provided. Design shall be such as to permit water to drain into the sump and not be trapped by baffles or otherwise stand in pockets. The sump shall have a bolted flange cover. A 1-inch, quick-close type, gate valve shall be provided in the bottom of the sump cover. The outlet of this valve and the control handle shall be modified so that a low point sample can be taken by operating personnel when standing along side of the vehicle. Baffles shall be provided and shall conform to ICC Regulations MC-303. The baffles shall be provided with at least 20-inch openings to permit access by maintenance personnel to all sections of the tank interior. The tank shall be designed so that all portions are accessible for cleaning and general maintenance.

3.6.2 Manhole.- The tank shall have only one top fill opening which shall be incorporated in a 20-inch manhole located as near the top center of the tank as possible. It shall be provided with vacuum and pressure operated vent with a minimum effective opening of forty-four hundredths of a square inch (0.44 sq. in.). The fill opening shall be 10-inches in diameter and shall have a cover equipped with spring loaded quick release latches or blow-out plugs to provide additional emergency vapor release. Blow-out plugs, if used, shall incorporate fine mesh fire screen. The fill opening cover shall be designed to open on a 90 degree angle with a hinge facing the cab of the vehicle. The fill opening cover shall be of a design that will close and lock automatically when the vehicle is moved. A liquid level indicator shall be provided. It will be incorporated in the manhole and will designate the normal liquid capacity of the tank.

3.6.3 Tank Top Walkway.- A tank top walkway shall be provided. It shall extend the entire length of the tank and shall be at least 20 inches wide. It shall be constructed of expanded metal and be securely fastened at all edges so as to provide a safe walking surface for personnel working on the top of the tank.

3.6.4 Overturn Protection.- Tank top equipment, such as fill covers, piping connections, etc., are to be provided with suitable protection in the event the vehicle overturns. Overturn protection will not have closed ends or other construction features that will trap spilled product or water accumulations. Tank top drains through the interior of the tank are not permitted.

3.6.5 Fenders.- Contractor shall provide fenders over rear four wheels. They shall be designed so as to provide ample protection to the vehicle when operating in muddy areas.

3.6.6 Ladders.- Steps or ladders with hand rails shall be provided at the left, front side of the tank and at the right, rear side of the tank to allow easy access from the ground to the tank top walkway.

3.7 Rear Cabinet.- A rear cabinet shall be provided. It shall be of sufficient size and strength to have hose reels, electric motors, static reel,

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and necessary valves and fittings. Cabinet shall be so constructed as to prevent water from entering or splashing in when doors are closed. Provisions shall be made for draining cabinet and drains shall not allow fluid to fall on any part of the vehicle. All doors shall be adequately supported by flush continuous-type hinges containing $\frac{1}{4}$-inch minimum diameter brass rods. Doors shall be reinforced to prevent warping and/or bending. Flush-type hardware, cadmium plated or otherwise processed to prevent corrosion, shall be provided. Access doors or panels of sufficient size and number shall be provided to permit servicing of all equipment contained within the cabinet. Door latches shall be slam type. All handles shall be lift type.

3.8 Pumping System.-

3.8.1 Pump.- The pump shall be a self-priming centrifugal type or positive displacement type aircraft fuel dispensing pump. The capacity of pumping system shall be 300 GPM, 40 PSI normal nozzle pressure, with nozzle surge limited to 120 PSI or less, with 0.5 second shut-off at nozzle. All parts of the pump in contact with fuel shall be constructed of non-ferrous material. Pump shall be so positioned as to assure that no damage may occur from drive shaft or other moving parts.

3.8.1.1 Pump Seal Failure.- The pump shall be constructed and positioned in a manner that should a pump seal fail, fuel will spill on to the ground and shall not enter any gear box or transmission.

3.8.1.2 Pump Capability.- The pump shall be capable of transferring aviation fuel conforming to Specification MIL-G-5572 and/or MIL-F-5624 at any constant flow rate, adjusting from 0 to 300 GPM when drawing fuel from storage tank at a level of one foot below the center line of the pump inlet assembly and delivering fuel through both the single point nozzle and the over-wing nozzle with the nozzles elevated 20 feet above the base of the pump.

3.8.1.3 Power Take-Off Connection.- The pump shall be connected to the power take-off by means of a shaft. The shaft will incorporate a slip joint and at least two universal joints.

3.8.1.4 Power Take-Off Control.- Both engagement and disengagement of the power take-off shall be capable of being performed by the operator while standing on the ground at the control panel. Positive indication of the exact position of the power take-off shall be provided. The operation of the power take-off between the fully engaged and the fully disengaged positions shall include provisions which cause the engine to be retarded to idle while this operation is taking place. Engagement of power take-off shall not be possible unless parking brakes have been set. A plate made of non-corrosive metal bearing the following inscription shall be provided.

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NOTICE

"Tank emergency valve shall be open and hand brakes applied before power take-off control is shifted into gear."

The plate shall be located adjacent to the power take-off control.

3.8.1.5 Pumping Control Panel.- All pumping system controls shall be mounted on a single panel in position convenient to the operator when he is standing on the ground at the rear of the vehicle. These controls shall include but not be limited to, power take-off control, internal emergency valve remote control, pressure gauges for filter/separator and fuel monitor engine tachometer and such other controls necessary to operate the pumping system.

3.9 Piping System.-

3.9.1 Pipe.- All pipe in contact with fuel shall be constructed of stainless steel Type 304L or Type 316. Pipe sections may be joined by welding or by means of split couplings.

3.9.1.1 Pipe Size.- Pipe size shall be compatible with other components of the system design and with the flow and pressure requirements spelled out in this specification. Light wall tubing is permitted. Bends and turns will be held to a minimum in order to reduce pressure loss.

3.9.2 Fittings.- Fittings such as elbows, tee's, crosses, laterals, wye's, etc., shall be fabricated of stainless steel type 314 or type 316.

3.9.3 Valves.- Manual valves may be constructed of brass or stainless steel. Automatic valves shall be constructed of aluminium. No aluminium valve shall be connected to another component of the system by thread. Flange or groove connections employing split couplings shall be used. Where a threaded valve is used a thread to groove adapter shall be used to connect to pipe. No pipe section shall be welded directly to a valve. All valves shall have a metal tag attached. The tag shall be etched or embossed with a number identifying the valve.

3.9.4 Piping Diagram.- An etched or embossed metal plate of corrosion resistant material showing the piping diagram and instructions for operating the pumping system shall be attached inside the rear cabinet at a point where it can be readily referred to. Identifying numbers of valves shall be shown on diagram.

3.9.5 Emergency Valve.- The tank shall be equipped with a flange mounted 4-inch elevated seat emergency valve. Valve seat to be approximately $2\frac{1}{2}$ to 3 inches above the bottom of the tank and at least 3 feet away from the water sump. The valve shall be equipped with a dead-man control. Dead-man control shall be air operated. Dead-man control shall be equipped with an air hose 25 feet in length. Hose to be contained on a spring loaded reel. A stop shall be provided to prevent damage to control valve should the hose be rewound too rapidly.

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In addition the valve shall be equipped with two emergency shut-off controls. One to be located at the right rear of the vehicle the other to be located at the left front of the tank. Either to be readily available to the equipment operator when he is standing on the ground. Each emergency shut-off control to have a sing painted adjacent to it in 3 inch letters. The sign to read "Emergency Valve Shut-Off". Method of operation shall be shown by the wor "Push" or "Pull" as appropriate. Emergency valve or its controls to incorporate fusible links to provide automatic closing of the valve in the event of fire.

3.10 Filter/Separator.- A filter/separator constructed from aluminum in accordance with MIL-F-00254 D dated 10 November 1965, Class B, Type 4-350 gpm shall be provided. Air elimination from the filter/separator shall be accomplished by providing a $\frac{1}{2}$ inch tube connection from the top of the filter/separator to the main fuel tank. The air bleed line shall be routed to the main tank in such a way so that the bleed fluid always enters below the liquid level in the tank. A check valve shall be incorporated in the bleed line to prevent back flow of fluid. Construction shall be in accordance with ASME Code for unfired pressure vessels, label shall be required.

3.10.1 Filter/Separator Pressure Gauge.- A pressure gauge with a $4\frac{1}{2}$ inch dial, 0-160-psi range, with figure intervals every 20 pounds and graduating marks every 2 pounds, shall be provided. The gauge shall have a bourdon tube of bronze. The case shall be aluminum and designed for flush mounting. The gauge case shall be moisture proof, gasket shall be neoprene. A means of preventing moisture accumulation on the inside of the gauge glass shall be provided. The gauge shall be flush mounted on the control panel. A three way valve shall be provided and the valve, gauge and filter/separator shall be interconnected by copper tubing so that the pressure drop can be read across the filter/separator.

3.10.2 Filter/Coalescer Elements.- The filter/coalescer elements shall be constructed in accordance with specification MIL-F-52308. They shall meet all the requirements of specification MIL-F-8901 and shall be the product of a manufacturer who has qualified his elements at a laboratory acceptable to the Government.

3.11 Fuel Quality Monitor.- A 300 GPM fuel monitor, Go-No-Go type, shall be furnished. It shall meet all the requirements of specification MIL-M-81380 (AS). A pressure gauge identical to the one called for in paragraph 3.10.1 of this specification shall be provided. It shall be mounted on the control panel and interconnected to the fuel monitor through a three-way valve for purposes of reading pressure drop.

3.11.1 Surge Suppressor.- A surge suppressor shall be provided. It shall be installed in the piping system at a point that will provide maximum protection to the system and all components thereof in the event of the system is subjected to pressure surges at any time during a fueling or defueling or hose evacuation operation.

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3.12 Meter.- One meter with a maximum rating of at least 300-gallons per minutes shall be housed in rear compartment. Meter shall be capable of monentary over-speeding to 125 percent of maximum rated capacity without mechanical damage or impairment of accuracy. Meter shall be designed for 75 pounds per square inch working pressure and 125 pounds per square inch hydrostatic test pressure. Meter shall have an accuracy in accordance with the requirements of Bureau of Standards Handbook No. H-44. Meter shall be capable of being adjusted, while under pressure, without leakage or loss of product. The adjustment sensitivity shall be sufficiently fine to permit calibration changes in conformance with accuracy requirements set forth above. The adjustment range shall be suitable for both aviation gasoline and JP-5 fuel and shall be capable of accomplishment without the use of external change gears. A vertical meter counter shall have a seven (7) figure totalizer and a four (4) or five (5) figure setback indicating full gallons. Figures in setback shall be coated with light reflecting material. All components of the meter in contact with fuel shall be fabricated from non-ferrous or stainless steel material.

3.13 Hose Reels.- Two hose reels shall be provided. One reel shall be capable of containing 50 feet of $2\frac{1}{2}$ inch collapsible hose and the other reel shall be capable of containing 50 feet of $1\frac{1}{2}$ inch non-collapsible hose. The reels shall have a gasoline-tight rotating joint (swing joint) of a commercial type. Such rotating joint shall have ball bearings and sealing that provides resistance to aromatic fuels at temperatures from -65°F to +125°F. The rotating joint shall not bind under loading and shall have provision for lubrication of moving parts and be capable of easy disassembly for maintenance purposes. Hose reel furnished for $1\frac{1}{2}$ inch hose shall have rotating joint and piping of 2-inch I. D. size. Hose reel furnished for $2\frac{1}{2}$ inch hose shall have rotating joint and piping of 3-inch I. D. size. The hose discharge connection of both reels shall have bolted flange connections and be replaceable in the event of thread damage. Provisions shall be made for lubrication of all working parts, such as bearings or other mating surfaces. All such parts shall be lubricated prior to delivery with a lubricant suitable in fuel service at temperatures ranging from -65°F to +125°F. Bearing seals will be such that pressure due to forced lubrication will not bind the shaft. Material in contact with fuel shall be non-ferric or stainless steel. In the event that both aluminum and bronze are used, they shall not be mated together unless a protective coating of zinc chromate is used. Piping shall be connected by flame rather than by thread. The hose reel elbow outlet shall be so arranged as to permit the hose to lie smoothly on the drum without kinking and to make possible a smooth second layer of hose. Hose rollers or formed supports shall be provided to prevent hose from wearing against sharp edges. Hose reel shall have anti-friction bearings on the side of the reel opposite the swing joint. Reels shall be provided with a drag brake. The reels may be gear or chain drive and shall operate at a speed of 43 RPM plus or minus 2 RPM during rewinding. One electric explosion-proof hose reel motor shall be installed on each hose reel for rewinding the hose. Voltage of motors shall be compatible with the vehicle electric system. Radio suppression shall be provided. Push button controls of the dead-man type shall be provided for the hose reel rewind motors. The push button controls shall be explosion-proof construction with a rubber

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boot covering the push button. It shall be of a type approved by the Underwriter Laboratories. A limit stop shall be provided to automatically stop the reel at the fully unwound position. All parts of the reel requiring servicing shall be easily disassembled and designed for easy access by standard tools.

3.14 Bottom Loading.- A bottom loading system shall be provided. It shall be so designed as to permit loading of the tank from the bottom without the requirement for a man on top of the truck. The system will be completely self contained on the vehicle, requiring no external electrical cables or hydraulic lines for operation, with the exception of the product supply line. Requirements for maintenance and replacement of working parts shall be held to a minimum.

3.14.1 System type.- The type system used shall eliminate the necessity for a float and associate mechanical linkage that might conceivably malfunction and result in an overfill of the tank. Systems employing electric energy to operate the shut-off mechanism are not permitted.

3.14.2 Shut-Off Valve.- The shut-off valve shall be a line mounted, balanced piston type. It shall be installed on the vehicle. Inlet and outlet of the valve shall be flanged. The valve shall be capable of withstanding a minimum static pressure of 150 PSIG. Operating pressure shall be from 10 PSIG to 100 PSIG. The valve shall be capable of a soft, shockless but positive shut-off. Pressure drops across the valve shall not exceed 8 PSIG at 600 GPM. All fittings shall be 3/8 inch O D flared tube type. A pressure regulator or back pressure valve shall be provided. The valve shall have a "dead-man" control feature to allow for immediate shut-off in case of emergency.

3.14.3 Precheck Valve.- A precheck valve shall be provided for purpose of checking the integrity of the bottom load system. It shall be a spring loaded, normally shut, lever operated type.

3.14.4 Loading Adapter.- The bottom loading adapter shall be a dry break type, equipped with an International Standard $2\frac{1}{2}$ inch bayonet locking flange type. The inlet flange shall be equipped with a product selection ring, with a minimum of six possible positions. Position numbers will be specified and will follow the format prescribed by the American Petroleum Institute. The adapter shall be compatible with the single point nozzle specified in this specification.

3.15 Defueling.- The unit shall be so designed as to permit the defueling of aircraft utilizing the vehicle pumping system. Piping shall be so arranged that the fuel being removed from the aircraft passes through the filter/separator before being returned to the vehicle tank.

3.16 Hose Evacuating.- The unit shall be capable of evacuating the $2\frac{1}{2}$ inch hose after an aircraft fueling has been completed. Essentially all fuel shall be removed from the hose. The evacuation system shall be so designed that all fuel furnished the eductor or other device shall be obtained from a source down stream of the filter/separator.

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3.17 Static Grounding Cable.- A static grounding cable shall be provided. It shall consist of at least No. 12 stranded wire, clear plastic covered, 80 feet in length. The cable shall terminate in a "Y" end. Each leg of the "Y" to be 15 feet. A heavy duty alligator clamp shall be attached to each of the loose ends of the "Y". The alligator clamps shall be constructed of copper alloy metal. The cable shall be contained on a spring loaded, interrupted gear, rachet type, with roller guides, reel. A cable stop which will prevent damage to the clamps because of rapid rewind will be provided.

3.18 Hose.-

3.18.1 Non-collapsible type.- Each refueler shall be equipped with one 50 foot length of $1\frac{1}{2}$ inch non-collapsible hose conforming to Specification MIL-H-6615.

3.18.2 Collapsible Hose.- Each refueler shall be equipped with one 50 foot length of $2\frac{1}{2}$ inch collapsible hose conforming to Specification MIL-H-22279.

3.19 Nozzles.-

3.19.1 Nozzle, Pressure Fuel Servicing, Type D-1.- Each refueler shall be equipped with a pressure fuel servicing nozzle, type D-1. It shall conform to Specification MIL-N-5877.

3.19.2 Over Wing Nozzle.- Each refueler shall be equipped with an overwing $1\frac{1}{2}$ inch fuel nozzle with a $1\frac{1}{4}$ inch spout. The nozzle shall incorporate venting flutes. The nozzle shall conform to Specification MIL-N-4180.

3.19.3 Single Point Nozzle Adapter.- A fueling spout adapter for use with the single point nozzle shall be provided. It shall be capable of permitting a flow of at least 150 GPM when used in conjunction with the single point nozzle. The design of the adapter shall be such as to preclude accidental detachment from the single point nozzle when in use in an over wing fueling operation. Appropriate provisions for stowing the adapter, when not in use, shall be provided on the vehicle.

3.20 Coupling Assembly, Quick Disconnect.- The $2\frac{1}{2}$ inch hose shall be attached to the pressure fuel servicing nozzle by means of a coupling assembly, quick disconnect. The coupling assembly shall conform to Specification MIL-C-19179. It shall be Class I as described by a fore-mentioned specification.

3.21 Cabinet Lights.- One or more as required vapor-proof, shatter-proof, lights shall be provided to illuminate rear cabinet interior. The light or lights shall be so positioned as to illuminate all instruments installed in rear cabinet. Voltage of lights shall conform to chassis voltage.

3.22 Circuit Breakers.- All electric circuits shall be protected with manual reset circuit breakers.

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3.23 Fire Extinguishers.- Two 30-pound capacity dury chemical fire extinguishers in accordance with specification MIL-E-22246 shall be provided. They shall be mounted on brackets located on the vehicle at points where they would be readily available in the event of fire. Brackets shall be of such a design as to permit release of extinguisher in one operation with one hand by personnel standing on the ground.

3.24 Bonding.- All metal parts of components shall be specifically bonded together to prevent possibility of spark due to static electricity. A rubber static drag strap shall be provided and located near the front axle. It shall be of sufficient length to allow at least four (4) inches of the strap to be in contact with the ground.

3.25 Tachometer.- An electric type engine tachometer shall be provided. It shall be mounted on the control console in such a manner as to be visible when operating the engine throttle control. The lens of the tachometer shall be of shatter proof material that is resistant to discoloration. A cork and neoprene gasket or other suitable material shall be used to insure that the dial is substantially weather proof.

3.26 Manual Container.- A waterproof container suitable for storage of Instruction Manuals shall be attached to an accessible area of the truck cab. The container shall be constructed of heavy duty canvas. The canvas shall be at least 10 ounce material, olive drab in color, with a flap cover. The flap cover shall be secured with snap fasteners. Overall dimensions shall be approximately 12 inches by 14 inches. It shall be labeled "Instruction Manuals."

3.27 Towing, Lifting and Tie-down Devices.- Towing, and tie-down devices shall be furnished and where practical these devices shall be combined. Towing shackles shall be substantially mounted on front of unit. Tie-down devices shall be suitable for securing the vehicle during transportation by aircraft, railroad or ship.

3.28 Name Plate.- The contractor shall furnish and affix a corrosion resistant metal name plate in accordance with standard MIL-STD-130 to the vehicle tank structure. It shall be located on the right side, as near the front of the vehicle as practicable. The name plate shall contain the following information:

Refueling Unit, Truck Mounted, Specification MIL-R- 81394
Truck Manufacturer _____
Model No. _____
Serial No. _____
Refueler Manufacturer _____
Unit No. _____
Delivery Date _____
Contract No. _____
USN Registration No. _____
Payload Allowance _____
Gross Vehicle Weight _____

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3.29 Finish.-

3.29.1 Treating and Painting.- All exterior surfaces which are normally painted in commercial practice shall be painted with not less than one coat of primer in accordance with Federal Specification TT-P-645 and two finish coats of semi-gloss enamel applied. It shall be in accordance with MIL-STD-193. Color shall be Marine Corps Green No. 23 in accordance with Federal Specification TT-E-485.

3.30 U. S. Navy Registration Numbers.- U. S. Navy Registration numbers shall be applied on the vehicles in accordance with Part 5, Section 1, of NAVDOCKS P-300.

3.30.1 Request for Registration Numbers.- Registration numbers shall be requested from Contracting Officer at least four (4) weeks prior to the anticipated shipping date of any vehicles.

3.31 Center of Gravity.- The contractor shall determine the center of gravity of the entire vehicle with tank empty and indicate by painting a vertical stripe one (1) inch wide and twelve (12) inches long, surmounted by the letters "C/G". Color of letters and characters shall be black and approximately one (1) inch high.

3.32 Vehicle Capacity.- The tank capacity in gallons, the limit dimension of the total vehicle in feet and inches, length, height, width, the total cube in cubic feet and the empty weight of the total vehicle shall be determined and stenciled on each side of the vehicle tank. Letters shall be black and approximately one (1) inch high.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality Program.- The contractor shall provide and maintain a quality program acceptable to the Government, for supplies and services covered by this specification. The quality program shall be in accordance with Military Specification MIL-Q-9858.

4.2 Supplier's Responsibility.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own or any other inspection facility and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections or tests set forth in the specification where such inspections or tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2.1 Notification.- The contractor shall notify the Government Inspector and the Commander, Naval Air Systems Command in sufficient time, prior to presenting the preproduction model for testing so that the Government representatives may witness all tests. All test equipment necessary for conducting tests in accordance with this specification shall be supplied by the contractor.

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4.3 Classification of Tests.- The inspection and testing of the refuelers shall be classified as follows:

(a) Preproduction Tests. . . . . See Paragraph 4.4
(b) Acceptance Tests. . . . . See Paragraph 4.5

4.3.1 Part Failure.- If during test a part fails, that part shall be replaced with a part of different design and/or material unless the procuring agency authorizes the installation of a new part of the original design and/or material. At the discretion of the procuring agency, redesigning and retesting may be required of any part which fails or indicated weakness during test but which was retained in the refueling unit to complete testing of other parts.

4.4 Preproduction Tests.-

4.4.1 Preproduction Model.- The preproduction model is defined as a production model which the manufacturer will subsequently produce in fulfillment of the contract. As soon as practicable after the award of the contract and prior to submission of any complete unit for final acceptance, the contractor shall furnish a preproduction model for examination and test to determine conformance to this specification. Approval of the preproduction model by the Government shall not relieve the contractor of his obligation to supply equipment conforming to this specification. Any change to or deviation from the preproduction model shall be subject to approval of the contracting officer.

4.4.2 Apparatus.- In so far as is practicable, apparatus used in conjunction with testing specified herein shall be of laboratory precision type and shall have been calibrated within the three (3) months immediately preceeding date of test. A certificate from the laboratory and/or person doing the calibration will be required. All data concerning details of this calibration shall be furnished the government inspector. All pressure readings on the preproduction model shall be taken by using transducers and recording type oscillograph having a capacity of at least six (6) channels. All transducers are to be individually calibrated immediately before tests are run. Copies of tracers from the recording of the oscillograph are to be furnished with written test report.

4.4.3 Liquid Flow Rates.- Liquid flow rates shall be determined in accordance with ASME, API Code for Distillation, Proving and Operation of Positive Displacement meters in liquid hydrocarbon service.

4.4.4 Test Fluid.- The test fluid used for all hydrostatic tests and for all liquid flow tests shall be stoddards solvent.

4.4.5 Data.- During testing of the pumping system, at least the following data shall be recorded:

Pressure at pump inlet - "HQ."
Pump discharge pressure - PSI
Pressure loss through meter - PSI
Pressure loss through filter/separator - PSI

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Pressure loss through hose - PSI
Pressure loss through nozzle - PSI
Pressure loss through fuel monitor - PSI
Slug control operation
Fuel flow rates - GPM
Engine oil temperature - ^{o}F
Engine speed - RPM
Pump speed - RPM

4.4.6 Detail Examination.- The chassis, tank and all components of the refueler shall be examined to determine performance to requirements of this specification in general and specifically as outlined below:

(a) Dimensional.

(1) Dimensions and capacities as required by specification.

(b) Functional.

(1) Presence of all parts, components and accessories as called for by requirements.

(2) Completeness and correctness of assembly.

(c) Materials.

(1) Suitability of materials for the purposes intended.

(d) Workmanship.

(1) Treatment and painting, incomplete preparation before painting. Such defects as scale, bumps, runs, thin spots, color not as specified or uneven.

(2) Suitability of fit at joints, seams, edges, etc., misalignment.

(3) Poor appearance of welds, porousness, flux inclusions, excessive spatter, under cutting, lack of fusions with parent metal or between parts.

(4) Dirty components.

(5) Completeness and adequacy of fastenings.

(e) Presence and acceptability of attachments, accessories, identification and other required markings, operating and safety instructions.

4.4.7 Cleaning.- Prior to the beginning of any tests, all refuelers including the preproduction model shall be subjected to the following cleaning procedure:

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Each refueler shall be carefully inspected for evidence of contaminants in the tank and fuel system such as loose, spattered or excess solder, metal chips or other debris. Stoddards solvent shall be pumped through the system for a minimum of ten (10) minutes at 100 GPM. At the end of the ten (10) minutes, strainers and low point drains shall be examined for evidence of contaminants. If any trace of contaminants is found, the recirculation of cleaning solvent at 100 GPM will continue for ten (10) minute periods until all trace of contaminants has been removed. Strainers and low point drains will be examined at the end of each ten (10) minute period.

4.4.8 Braking Test.- The fully loaded truck will be brought to a complete stop while traveling at twenty (20) MPH. Brakes will be applied in a rapid manner in order to effect a sudden stop. There shall be no failure of parts or materials furnished by the contractor as a result of this test.

4.4.9 Tank Air Test.- The 2000 gallon tank shall be subjected to a static air pressure test at three (3) psi for a period not less than 30 minutes. A soap solution shall be applied on the exterior of the tank at all weld and joints to facilitate the detection of any leaks.

4.4.10 Piping System Hydrostatic Test.- All components of the piping system including the internal valve, all pipe fittings, hose, nozzles and other components through which fuel passes shall be hydrostatically tested at a pressure of 125 psi for a period of at least thirty (30) minutes. The test fluid shall be stoddards solvent. There shall be leaks.

4.4.11 Vent Operation Test.- The venting equipment shall be removed from the tank, installed in a test fixture and bench tested. The vents shall relieve positive and negative pressures and prevent leakage or spillage of fuel in any position in accordance with the detail requirements of this specification.

4.4.12 Flow Test.- Stoddards solvent at normal ambient temperature shall be pumped from the tank to a point which is twenty (20) feet above the pump base. Flow rates shall be as follows, 50 GPM, 100 GPM and 300 GPM. Each flow rate shall be for a period of minimum of ten (10) minutes. Flow shall be through the single point nozzle. Nozzle inlet pressure shall not exceed 45 PSI. Following this, a second flow test shall be performed using $1\frac{1}{2}$ inch hose and the over-wing nozzle. Flow rates shall be 50 and 100 GPM for a minimum of ten (10) minutes at each flow rate. In all flow test, a suitable shut-off valve may be installed at the outlet of the nozzle to simulate the 20 feet head if desired. Maximum vacuum at the pump suction shall be 7 inches of mercury. During flow tests recording of data shall be in accordance with Paragraph 4.4.5 of this specification.

4.4.13 Bottom Loading System Test.- The vehicle tank shall be filled to its rated capacity through the bottom loading connection. The system shall be capable of shutting off flow automatically when the tank is fully loaded.

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4.4.14 Valve Closure Test.- While pumping at 300 GPM through the single point nozzle and with the nozzle raised a minimum of ten (10) feet above ground level, at least ten (10) valve closures at 0.5 a second closure time shall be taken in the manner prescribed by Paragraph 4.4.2 of this specification. The total pressure build-up attthe nozzle, including the 45PSI, the 10-foot head and any surge pressure shall not exceed 120 PSI.

4.4.15 Hose Evacuation Tests.- Upon completion of the flow tests the hose evacuation system shall be tested. Substantially all fuel shall be removed from the $2\frac{1}{2}$ inch hose. A minimum of ten (10) evacuation tests shall be made.

4.4.16 Pump Suction.- The pump shall be capable of producing a minimum vacuum of eighteen (18) inches of mercury when loading the truck from an outside source.

4.4.17 Hose Reel Tests.- Fluid handling sections of hose reels shall be given hydrostatic tests of 200 PSI to determine strength and porosity. Stoddards solvent shall be used as a test fluid. Pressure shall be applied in fifty (50) PSI increments. Pressure will be held for ten (10) minutes at each increment. There shall be no leaks. The hose reels shall be subjected to a vacuum test of at least 22 inches of mercury for a minimum of ten (10) minutes. There shall be no leaks.

4.4.17.1 Hose Reel Strength Test.- Hoses will be completely unwound and laid out at an angle of 45° from the axis of the vehicle. The hose shall be rewound a minimum of twenty (20) times. The 45° angle will be maintained during rewind as much as possible in order to bring hose to bear on the reel discs. There shall be no evidence of bending of the reel discs.

4.4.18 Test Records.- Contractor's records of all tests and inspections giving the results of said tests and inspections to determine compliance with the requirements of this specification shall be kept complete and available to the Governments representative at all times. Test and inspection records shall be signed and approved by a person specifically assigned by the contractor. Contractors not having laboratory testing facilities satisfactory to the government shall engage the services of a commercial testing laboratory capable of condutting required tests to determine compliance with all the requirements of this specification.

4.5 Acceptance Tests.- Each refueler of the production run shall be subjected to all of the requirements of the following paragraphs of this specification: 4.4.6, 4.4.7, 4.4.9, 4.4.10, 4.4.11, 4.4.13. In addition a modified flow test shall be performed to determine that all parts are functioning properly. A minimum of 2000 gallons of test product shall be circulated through the system at 300 GPM using both hoses. Prior to start of this flow test, elements shall be removed from both the filter/separator and the fuel monitor. At the end of the flow test, all test product shall be completely drained from the tank, filter/separator and all fluid handling piping and components. Elements shall be reinstalled in filter/separator and the fuel monitor.

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4.6 In-Service Evaluation.- There is hereby created an option in the Government to conduct an extended in-service evaluation test of four weeks duration on the preproduction model at a government facility following factory testing of the prototype vehicle and prior to the commencement of work on any of the production units. Shipment to the Government facility and return of the vehicle to the contractor's plant shall be on a Government bill of lading. The Government will provide an operator for the vehicle and such other personnel necessary to complete the in-service evaluation. Contractor personnel may participate in the in-service evaluation if desired. Any contractor costs incident to such participation will be at contractor's expense. During the in-service evaluation the vehicle will be subject to the following tests:

(a) Driven a minimum of 500 miles over improved hard surfaced roads at speeds varying from five (5) to thirty-five (35) miles per hour.

(b) Driven a minimum of 200 miles over unimproved gravel roads at speeds varying from five (5) to twenty-five (25) miles per hour.

(c) Driven off road and over rough fields at a speed of ten (10) miles per hour for a distance of twenty (20) miles.

(d) Driven through water during a fording operation. Water depth to be a maximum of five (5) feet. Speed to be four (4) miles per hour.

During all the above, the truck tank will be loaded with 1000 gallons of fuel. Following the road tests and the off road test, the vehicle will be used to fuel Naval and Marine Corps aircraft. Fuel operations will be performed by enlisted personnel employing standard fueling procedures established by Naval Air Systems Command directives.

There shall be no failure of any part or component furnished by the contractor as part of the completed vehicle and in conformance with this specification and the contract.

5. PREPARATION FOR DELIVERY

5.1 Packing for Shipment.- Packing for shipment shall be performed by the contractor in such a manner as to insure acceptance by common or other carriers for safe transportation, at lowest rate to the point of delivery dcalled for of contract material. Loose components shall be secured in the cab of the vehicle. Markings shall be in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended Use.- The refueling units covered by this specification are intended for use at advance base airfields. They will be operated over improved and unimproved roads. In addition, they will on occasion be operated off road over rough fields. They will be required to operate in water of a maximum depth of five (5) feet during fording of rivers and streams

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and when being taken ashore from landing craft. They will be required to operate in all kinds of weather and within temperature ranges common to the Continental limits of the United States.

6.2 Corrosion Resistance.- The term "Corrosion Resistant" as applied to various components of the refueling units covered by this specification is defined as "materials that shall resist corrosion while in contact with or exposed to highly aromatic fuels (115/145 octane aviation gasoline), water, air or any combination of these elements. While the degree of "resistance" is not specified in terms of penetration, it is understood that the definition of "resistance" will be that normally accepted in trade parlance.

6.3 Power Transmission Train.- The power transmission train, including the pump, must be adequate for the purpose intended and must have demonstrated thorough in-service operation, ability to sustain operating requirements without failure of any component of said transmission train.

6.4 Requirement for Descriptive Literature.- Descriptive literature as specified in the Invitation for Bids must be furnished as a part of the bid and must be received before the time set for opening bids. The literature furnished must identify the item in the bid to which it pertains. The descriptive literature is required to establish, for the purpose of bid evaluation and award, details of the products the bidder proposes to furnish as to the following items:

(1) Furnish three (3) flat drawings to scale with dimensions showing left side view, right side view, top view, front and rear views identical to the product the bidder proposes to furnish.

(2) Furnish make, model and rated capacity of the refueling pump.

(3) Furnish make, model and rated capacity of the power take-off.

(4) Furnish make, model and rated capacity of the filter/separator.

(5) Furnish make, model and rated capacity of the fuel monitor.

(6) Furnish make, model and rated capacity of the meter.

(7) Furnish make, model and rated capacity of the hose reels.

(8) Furnish make and model of bottom loading system.

(9) Furnish make and model of overwing nozzle.

(10) Furnish make and model of single point nozzle.

(11) Furnish a complete description of, including make, model and type, of the instruments that will be used in testing of all components during the preproduction tests.

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Failure of descriptive literature to show that the products offered conform to the specifications and other requirements of the Invitation for Bids will require rejection of the Bid. Failure to furnish the descriptive literature by the time specified in the Invitation for Bids will require the rejection of the Bid, except that if the material is transmitted by mail and is received late, it may be considered under the provisions for considering late bids, as set forth in the Invitation for Bids.

In addition to the required number of copies of the above descriptive literature, one (1) additional copy shall be furnished with each bid for retention by the Naval Air Systems Command.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-R-81394(AS) REFUELING UNIT, AIRCRAFT, 2000 GALLON TYPE MC-1A

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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